Lars Larsen

Hardware Product Development Lead: electronics, firmware, mechanical, manufacturing

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SUMMARY

Lead designer of every LZX Industries product since 2011: more than 100 video synthesizers, Eurorack modules, standalone instruments, kits, cases and power supplies. I designed some on my own and led a small team on others. My work covers analog and video circuits, FPGA design, embedded C++ firmware (8 years), sheet-metal enclosures and contract manufacturing. $6.4M in direct sales to customers in 50 countries.

EXPERIENCE

Founder & Principal Product Engineer, LZX Industries

Portland, OR · 2011–Present

Maker of video synthesizers and video processing instruments (DBA 2011, LLC 2017). Lead designer of every product. Some projects I designed on my own; on others I led a small team, with partners on firmware/HDL and board layout.

- Took more than 100 hardware products from concept to production across six Eurorack series, standalone instruments, DIY kits, cases and power. About 240 production circuit board designs and 140 metal front panels. $6.4M in direct sales (2018 to 2026), plus a dealer channel.
- Videomancer (shipping since late 2025): designed a standalone FPGA video synthesizer around an iCE40 HX4K and RP2040, with ADV7611/ADV7513 HDMI in and out, ADV7181C and ADA4412-3 analog input, and an ADV7393 SD/ED/HD encoder. Users load effects as Ed25519-signed packages. Released 69 public firmware release candidates and an open GPL-3.0 SDK.
- Embedded C++ since 2018: primary firmware author for TBC2 and Memory Palace (Zynq-7010, bare-metal and FreeRTOS), Scrolls (STM32F411), and Chromagnon and Videomancer (RP2040). Built the shared application engine (state machines, UI, presets, modulation, MIDI, USB) over three generations; the current C++20 version uses no exceptions, RTTI, STL containers or heap.
- Chromagnon (RevJ prototype on the bench): led a video-rate analog/digital instrument through seven board revisions to an 8-layer core. Replaced the Zynq SoM with an iCE40 and RP2040 controlling an analog Gilbert-cell XY engine, after costing an all-digital dual-ECP5 version. On-board silicon cost fell 64% ($195 to $70) and BOM cost a further 26%.
- Designed analog video circuits for shipped product lines: matched-pair multipliers and blenders, keyers, colorspace mapping, sync separation and video-rate op-amp stages. Co-defined the LZX 1V signal convention (output stages, 100 kΩ inputs) used in every Eurorack series.
- Gen3 power entry boards (FPGA12, SYNC12, PSU8, PSU16; shipped, revised 2021 to 2023): one +12 V input with diode-OR and EMI filter, a synchronous buck and an inverting converter to ±5.5 V, and RF LDOs down to clean ±5 V for analog video. Sequenced iCE40 rails and separately filtered clock and sync supplies. Used in about 15 Eurorack modules and TBC2.
- Mechanical: sheet-metal enclosures (Videomancer, made by HLH Prototypes; Eurorack cases and bus shields from SendCutSend), injection-molded custom knobs and about 140 metal front panels, modeled in Inventor and FreeCAD.
- Manufacturing: built the line through about 40 active suppliers (JLCPCB, HLH Prototypes, SendCutSend, DigiKey). Ran 1,281 manufacturing orders in MRPeasy, moved to InvenTree and ERPNext in 2026. Directed board layout by a partner.
- Software: led the Shopify Hydrogen/React storefront since 2023 (about 58k lines, Playwright tests). In 2026, using AI coding agents: a Python operations CLI (about 1,000 commands, 3,500 tests) linking Shopify, InvenTree, ERPNext, JLCPCB and UPS; a Rust/Tauri firmware updater; self-hosted Forgejo CI and Authentik SSO.

Operations Manager, Darkplace Manufacturing

Portland, OR · 2015–2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX, 2008–2010 and 2011–2015
- Lead Service Technician, Switched On (synthesizer sales and repair), Austin TX, 2010
- New Media Developer, FUNimation Entertainment, Flower Mound TX, 2007–2008
- Multimedia Developer, TargetScope, Plano TX, 2006–2007

SELECTED PROJECTS

- **Videomancer SDK (GPL-3.0)**: iCE40 video platform, program ABI, signed package format and GHDL image tester. About 190 C++ and 350 VHDL unit tests. github.com/lzxindustries/videomancer-sdk
- **Chromagnon web simulator**: Browser version of the Chromagnon instrument model (WebGL), checked against reference vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **LZX documentation**: Manuals, patch guides and specifications for the product line (open source, github.com/lzxindustries/lzxdocs). docs.lzxindustries.net
- **Cadet open hardware**: DIY video synth module kits, open-sourced in 2022 with schematics, layouts and BOMs. github.com/lzxindustries/lzxcadet
- **LZX Industries storefront**: Shopify Hydrogen/React storefront with product pages, firmware downloads and online tools. lzxindustries.net

SKILLS

Electronics	Analog, mixed-signal and video (SD to 1080p, composite, S-Video, YPbPr, RGB, HDMI, genlock), discrete high-speed analog, power entry and regulation, ngspice/SPICE, IBIS, DipTrace, Eagle, KiCad, DFM, BOM costing, board bring-up, oscilloscope automation
Firmware & FPGA	C/C++ (C++17/20, no heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing, VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40), Vivado (Zynq-7000)
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosures, injection-molded knobs, front panel graphics, Blender rendering
Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI coding tools (Claude Code, Cursor)
Product & operations	Concept to production, PLM (MRPeasy, InvenTree), ERP, supplier and contract manufacturer management, pricing, documentation, open-source hardware

EDUCATION

- B.A., Radio, Television & Film, University of North Texas, 2006